



Value at Risk of Options

Risk Management in Financial Institutions

Lecture 4_1

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The case of Barings



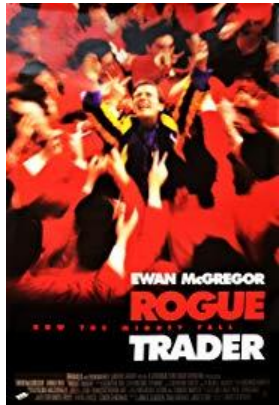
VaR relevant films 1: Wall Street (1987)

Michael Douglas as
Gordon Gekko (Academy
Award, Best Actor)
Directed by Oliver Stone



„The bottom line, ladies and gentlemen, is that greed is, for lack of a better word, a good thing. Greed is right. Greed is motivating. Greed leads to clarity, cuts through everything, sums up the essence of evolution. Greed in all its forms, greed for life, money, love or knowledge, drives humanity forward.”

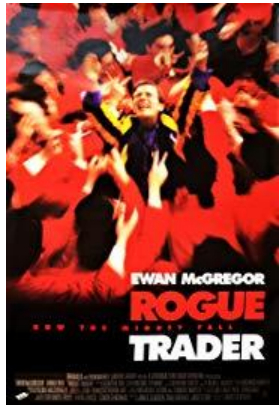
VaR relevant films 2: Rogue trader (1999)



Starring: Ewan McGregor as Nick Leeson



Rogue trader, 1999



**Starring: Ewan McGregor as
Nick Leeson**



The case of Barings



Barings:

- Over 200 years old English investment bank
- Trades on the Singapore International Stock Exchange (SIMEX) and the Osaka Stock Exchange (OSE)
- Predominantly proprietary trading
- Proprietary trading based on bank policy on an arbitrage basis only (risk-free profit making)



- The rationale behind the trading strategy: exploiting the relative mispricing of derivatives on the Nikkei 225 (Japanese stock index) between OSE and SIMEX
- Open overnight positions prohibited → intraday positions only
- Options trading prohibited (except on client order)

The case of Barings



- Nick Leeson has led futures trading at the recently founded Barings Securities Singapore (BSS) since 1992 (front@back office)
- He can trade on his own
- Basis of his trading strategy - 3 futures market instruments
 - Futures contracts on Nikkei 225
 - 10-year Japanese government bond futures (JGB futures)
 - Euroyen futures
- Nick Leeson's strategy initially successful → incredible profits
- Until...

The case of Barings

Long Nikkei 225 futures :
1992. July: 2 051 contracts
1995. January: 90 000 contracts
+
Short JGB futures:
1995. January: 28 000 contracts
+
Euroyen futures



1995. January: 55 399 contracts for Nikkei 225 with March maturity
5 640 contracts for the Nikkei 225 with a maturity in June

1995. January. 17.: Kobe earthquake



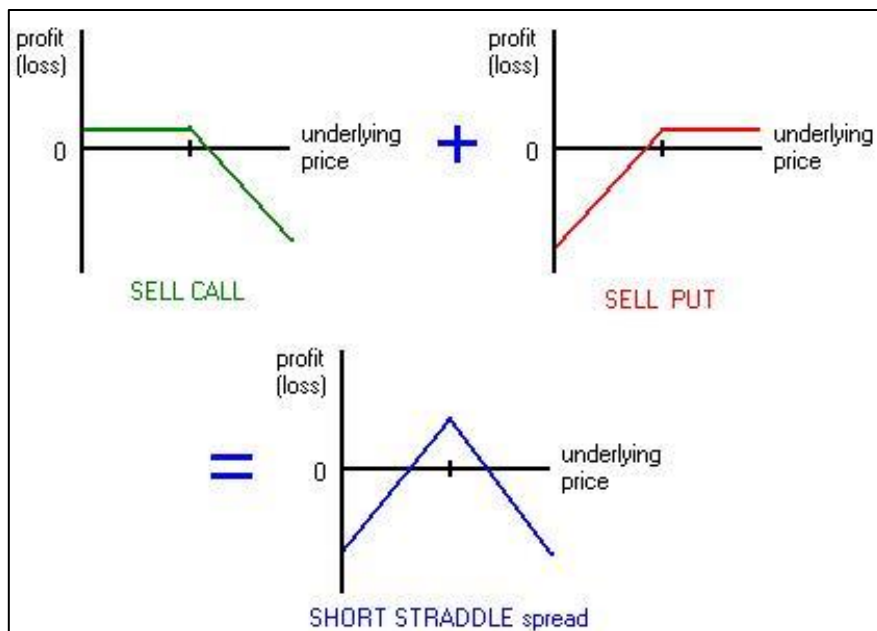
Nikkei 225 falls more than 1000 points



serious losses (early 1995: hundreds of millions of pounds)

The case of Barings - options

- When liquidity was scarce:
- Transaction to fund a margin requirement: option short straddle on the Nikkei 225
- close to 0 VaR at start!



Short Call + Short Put



Betting on the stability and constancy of the market



Sale of volatility



Profit: amount of option premiums paid by the buyer of options

The case of Barings

- Leeson creates a technical account No. 88888 in July 1992 to account for certain loss-making transactions
- This account is not seen in London, is not reported
- Thereafter, he exponentially increases the volume of options and futures trading through the account



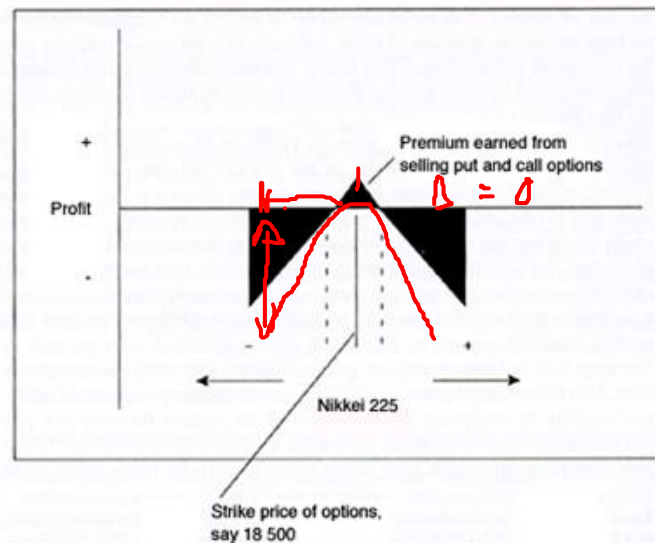
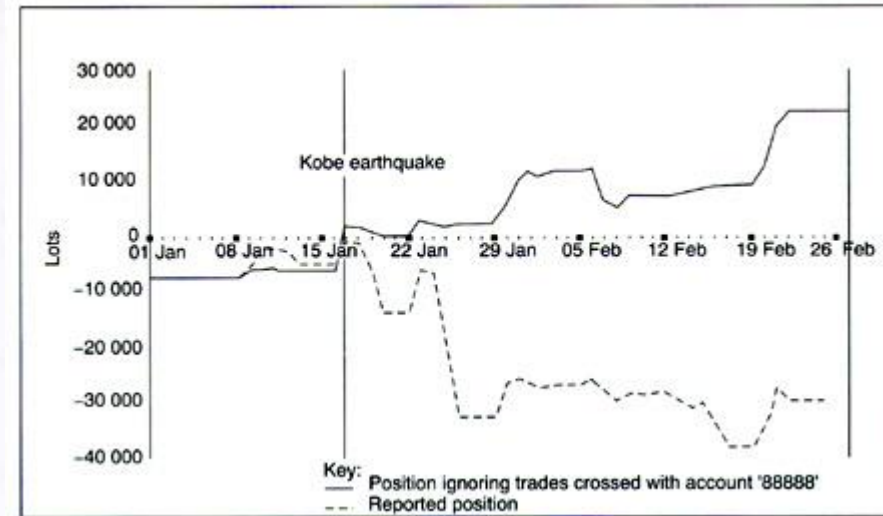
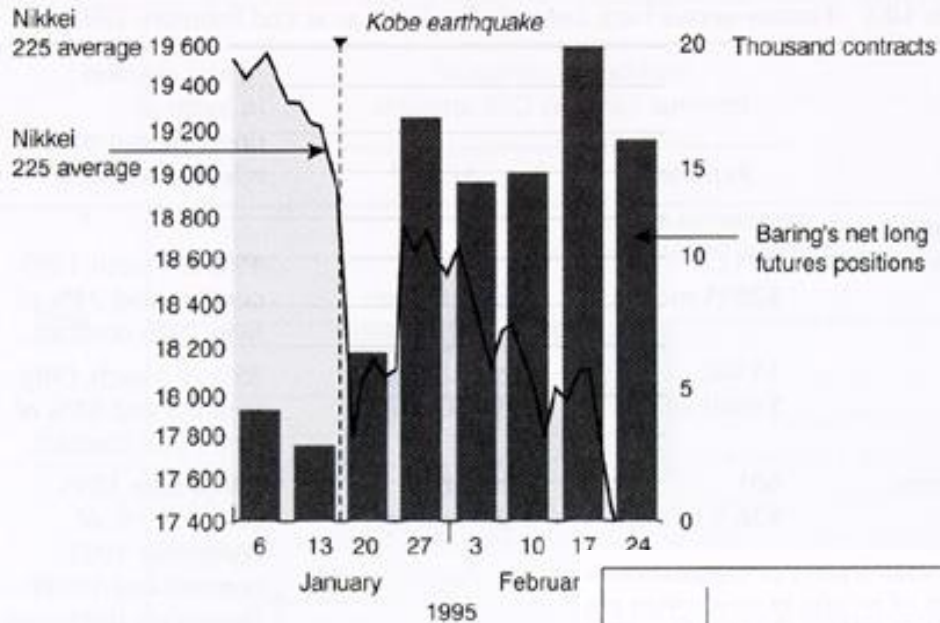
- Technical account No. 88888:
 - End of 1994: £208 million loss
 - Start of 1995: £830 million loss



- Barings equity reduced from \$615 million to \$0
- Bank of England sets example by letting the bank go bankrupt
- March 1995: Dutch ING acquires Barings for £1
- Nick Leeson: sentenced in Singapore to 6 and a half years in prison



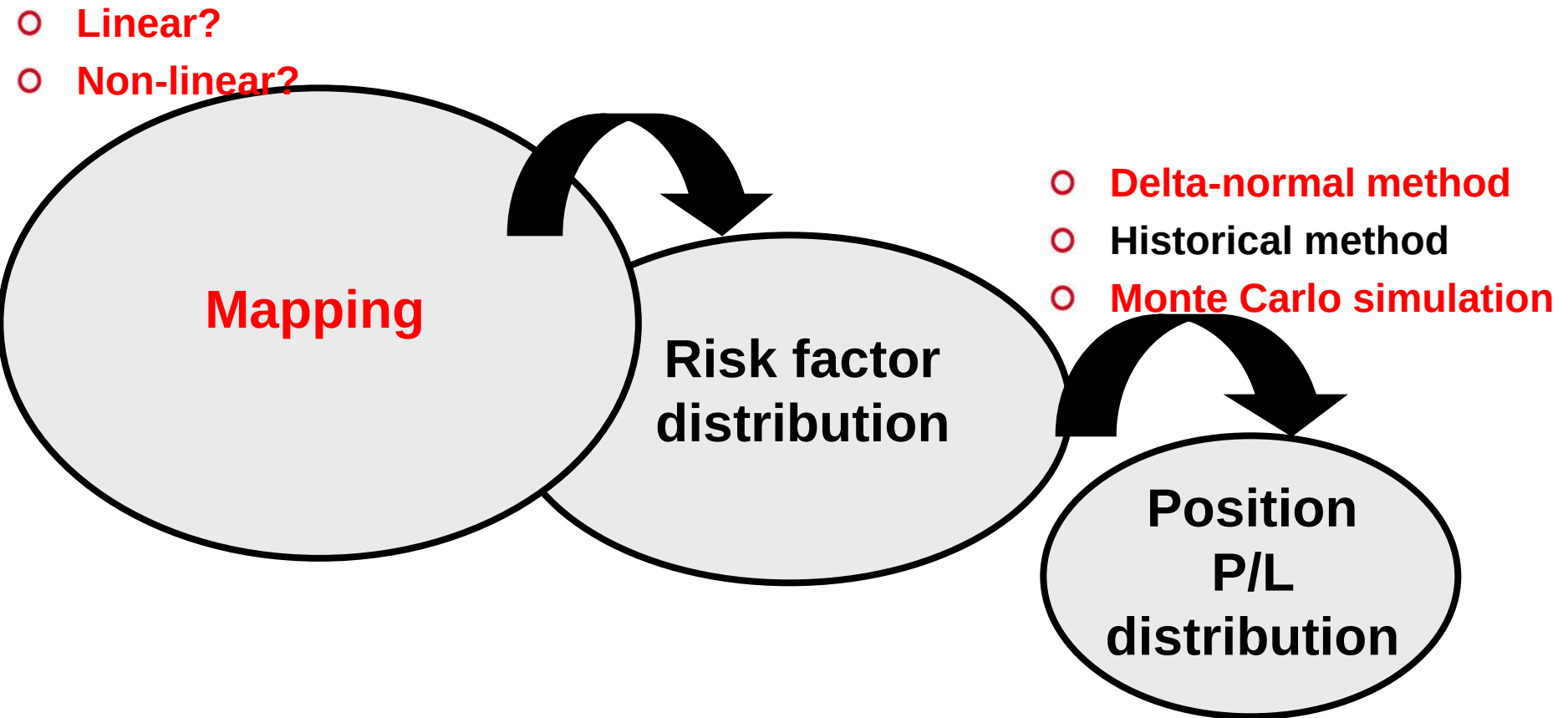
The case of Barings



Key:

----- break-even

Logical model for determining the Value at Risk



Mapping the position: the key to VAR is tracing back to the risk factor

5. Option position mapping

The arbitrage-free price of a long call option:

$$c = N(d1) \cdot S - N(d2) PV(X)$$

What should price changes be mapped to?



1. Full evaluation
(pl Black-Scholes)
 $S, r, \sigma, (T)$



Monte-Carlo simulation



2. „Greek letters”
 $S, S^2, r, \sigma, (T)$



Delta normal

Method based on full evaluation

$$c = c(S, \sigma, r, t) = N(d1) \cdot S - N(d2) PV(X)$$

- $d1 = \ln[S/PV(X)] / \sigma\sqrt{T} + 0.5\sigma\sqrt{T}$
- $d2 = d1 - \sigma\sqrt{T}$
- S : price of the underlying product
- σ : volatility
- r : interest rate
- T : time
- X : strike rate

Taylor Series Expansion (Equation 8.1, page 167)

$$\Delta P = \frac{\partial P}{\partial S} \Delta S + \frac{\partial P}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 P}{\partial S^2} (\Delta S)^2 + \frac{1}{2} \frac{\partial^2 P}{\partial t^2} (\Delta t)^2 + \frac{\partial^2 P}{\partial S \partial t} \Delta S \Delta t + \dots$$

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„Greek letters ” based mapping

$$dP = \Delta \cdot dS + \frac{1}{2}\Gamma \cdot dS^2 + \Lambda \cdot d\sigma + \rho \cdot dr + \Theta dt + \dots$$

- **S**: price of the underlying product
- **σ**: standard deviation of underlying (volatility)
- **r**: a risk-free interest rate with the same maturity as the option



Delta-normal VaR of options

$$VaR(dc) = 1,645 \cdot \sqrt{\Delta^2 \cdot S^2 \cdot \sigma^2 + \frac{1}{2} \left[\Gamma S^2 \cdot \sigma^2 \right]^2}$$

$$\Delta = N(d_1) \quad \Gamma = \frac{1}{\sigma \cdot S} \cdot \frac{N'(d_1)}{\sqrt{T}}$$

A Trader's Gold Portfolio. How Should Risks Be Hedged? (Table 8.1, page 154)

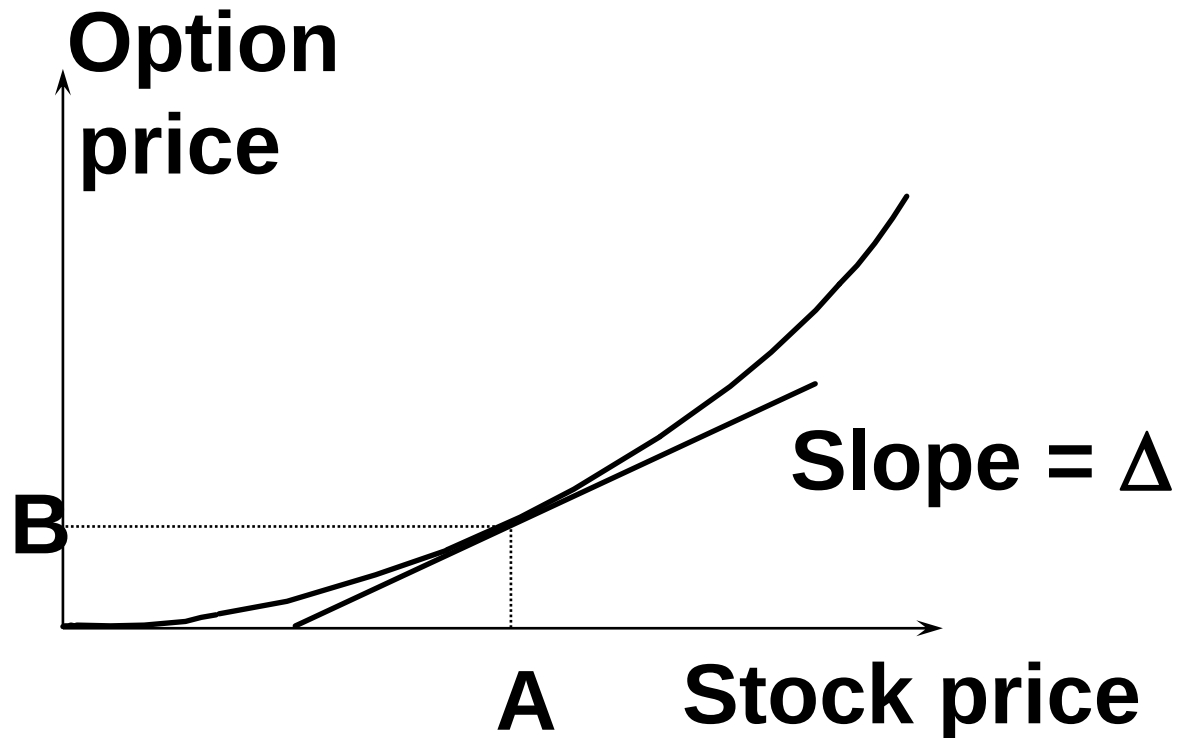
Position	Value (\$)
Spot Gold	9,180,000
Forward Contracts	– 3,060,000
Futures Contracts	2,000
Swaps	180,000
Options	– 6,110,000
Exotics	125,000
Total	317,000

Delta

- Delta of a portfolio is the partial derivative of a portfolio with respect to the price of the underlying asset (gold in this case)
- Suppose that a \$0.1 increase in the price of gold leads to the gold portfolio decreasing in value by \$100
- The delta of the portfolio is -1000
- The portfolio could be hedged against short-term changes in the price of gold by buying 1000 ounces of gold. This is known as making the portfolio delta neutral

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Delta of the Option



Linear vs Nonlinear Products

- When the price of a product is linearly dependent on the price of an underlying asset a “hedge and forget” strategy can be used
- Non-linear products require the hedge to be rebalanced to preserve delta neutrality

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Example of Hedging a Nonlinear Product (pages 156-160)

- A bank has sold for \$300,000 a European call option on 100,000 shares of a nondividend paying stock
- $S_0 = 49$, $K = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 20$ weeks, $\mu = 13\%$
- The Black-Scholes-Merton value of the option is \$240,000
- How does the bank hedge its risk to lock in a \$60,000 profit?

Delta Hedging

- Initially the delta of the option is 0.522
- The delta of the position is -52,200
- This means that 52,200 shares must be purchased to create a delta neutral position
- But, if a week later delta falls to 0.458, 6,400 shares must be sold to maintain delta neutrality
- Tables 8.2 and 8.3 (pages 158 and 159) provide examples of how delta hedging might work for the option.

Table 8.2: Option closes in the money

Week	Stock Price	Delta	Shares Purchased
0	49.00	0.522	52,200
1	48.12	0.458	(6,400)
2	47.37	0.400	(5,800)
3	50.25	0.596	19,600
....			
19	55.87	1.000	1,000
20	57.25	1.000	0

Table 8.3: Option closes out of the money

Week	Stock Price	Delta	Shares Purchased
0	49.00	0.522	52,200
1	49.75	0.568	4,600
2	52.00	0.705	13,700
3	50.00	0.579	(12,600)
....			
19	46.63	0.007	(17,600)
20	48.12	0.000	(700)

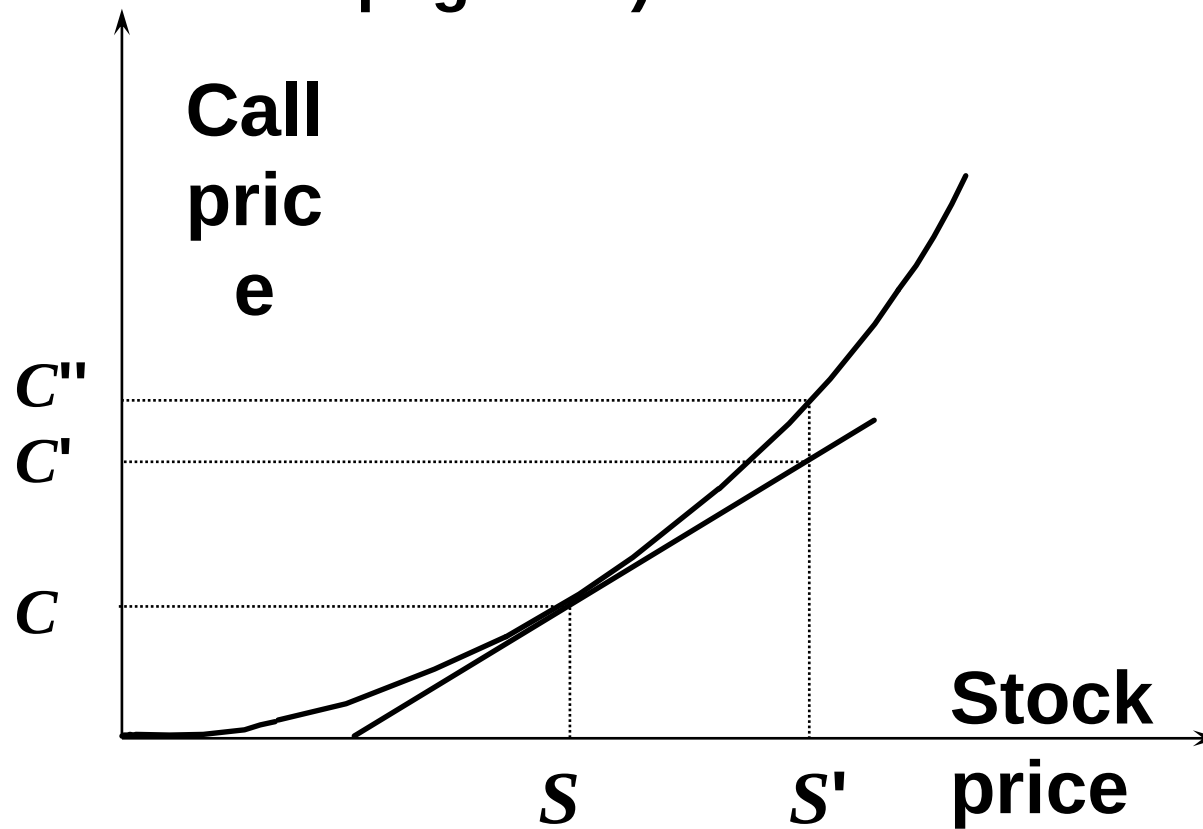
Gamma

- Gamma (Γ) is the rate of change of delta (Δ) with respect to the price of the underlying asset
- Gamma is greatest for options that are close to the money

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Gamma Measures the Delta Hedging Errors Caused By Curvature (Figure 8.4,

page 161)



Gamma and Vega Limits

- In practice, a traders must keep gamma and vega within limits set by risk management

The Linear VaR Model and Options

Consider a portfolio of options dependent on a single stock price, S . Define

and
$$\delta = \frac{\Delta P}{\Delta S}$$

$$\Delta x = \frac{\Delta S}{S}$$

Linear Model and Options continued

- As an approximation

$$\Delta P = \delta \Delta S = S \delta \Delta x$$

- Similarly when there are many underlying market variables

$$\Delta P = \sum_i S_i \delta_i \Delta x_i$$

where δ_i is the delta of the portfolio with respect to the i th asset

Example

- Consider an investment in options on Microsoft and AT&T. Suppose the stock prices are 120 and 30 respectively and the deltas of the portfolio with respect to the two stock prices are 1,000 and 20,000 respectively
- As an approximation

$$\Delta P = 120 \times 1,000 \Delta x_1 + 30 \times 20,000 \Delta x_2$$

where Δx_1 and Δx_2 are the percentage changes in the two stock prices

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But the Distribution of the Daily Return on an Option is not Normal

The linear model fails to capture skewness in the probability distribution of the portfolio value (*due to various Greek letters*).

Impact of Gamma

(See Figure 14.1, page 313)



**Positive
Gamma**

Negative Gamma

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Monte Carlo Simulation (page 316-317)

To calculate VaR using MC simulation we

- Value portfolio today
- Sample once from the multivariate distributions of the Δx_i
- Use the Δx_i to determine market variables at end of one day
- Revalue the portfolio at the end of day

Monte Carlo Simulation continued

- Calculate ΔP
- Repeat many times to build up a probability distribution for ΔP
- VaR is the appropriate fractile of the distribution times square root of N
- For example, with 1,000 trial the 1 percentile is the 10th worst case.

Alternative to Normal Distribution Assumption in Monte Carlo

- In a Monte Carlo simulation we can assume non-normal distributions for the x_i (e.g., a multivariate t -distribution)
- Can also use a Gaussian or other copula model in conjunction with empirical distributions



**Thank you for the
attention!**

